FEDERAL SPECIFICATION

TALCUM; POWDER, PLAIN AND BORATED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers two types of talcum powder for personal use.

1.2 Classification.

1.2.1 *Types.*—Talcum powder shall be of the following types as specified:

Type I.—Plain
Type II.—Borated

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.—The following specifications and standards of the issues in effect on date of invitation for bids form a part of this specification to the extent specified herein:

Federal Specifications:

PPP–C–96—Cans, Metal, 28 Gage and Lighter.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Standards:

MIL–STD–105 — Sampling Procedures and tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractor, in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. — The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids shall apply:

Department Of Health, Education And Welfare:

Federal Food, Drug and Cosmetic Act and General Regulations for Its Enforcement.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.)

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Official Methods of Analyses of the Association of Official Agricultural Chemists.

(Application for copies should be addressed to the Association of Official Agricultural Chemists, Post Office Box 540, Benjamin Franklin Station, Washington 25, D. C.)

The Pharmacopoeia of the United States.

(Application for copies should be addressed to the Mack Publishing Company, Easton, Pennsylvania.)

Scotts Standard Methods of Chemical Analyses.

(Application for copies should be addressed to the D. Van Nostrand Company, Inc., New York City, N. Y.)

Consolidated Freight Classification Rules.

(Application for copies should be addressed to the Consolidated Committee, 202 Chicago Union Station, Chicago 6, Ill.)

3. REQUIREMENTS

3.1 Preproduction sample.—When specified, before production is commenced, a sample of the finished talcum powder shall be submitted for approval of the contracting officer or his authorized representatives for inspection and tests as specified in section 4 to determine compliance with this specification. The approval of the preproduction sample authorizes the commencement of production, but does not relieve the contractor of responsibility for compliance with all requirements of this specification.

3.2 Materials.—All materials used in formulation of the talcum powder shall conform to the requirements of the Federal Food, Drug and Cosmetic Act and general regulations issued for its enforcement. The contractor shall furnish a certificate of compliance covering the foregoing requirement to the contracting officer.

3.3 Composition. — Talcum powder shall consist principally of finely powdered native magnesium silicate with the addition of small amounts of perfuming agents and other ingredients consistent with cosmetic industry practice. The finished talcum shall contain no starch or carbonates and shall conform to the following composition limitations:

3.3.1 *Magnesium content.* — The magnesium content, calculated as magnesium oxide (M_gO) shall be not less than 30 percent for type I and 26.5 percent for type II when tested as specified in 4.4.3.

3.3.2 *Toxic metals.*—The maximum content of lead (as Pb) and arsenic (as As_2O_3) shall be 20 parts per million and 2 parts per million respectively when tested as specified in 4.4.4 and 4.4.5.

3.3.3 *Type II (Borated).*—Borated talcum powder shall contain the following additional ingredients in the amounts specified on the basis of percent by weight of the finished formulation:

	Minimum	*Maximum*
Boric acid	2.0	6.0
Zinc stearate	1.0	5.0

3.3.4 *pH.*—A water suspension of the talcum powder shall be neutral or slightly acidic as applicable to litmus paper.

3.3.5 *Loss on ignition.*—The maximum loss permitted on ignition shall be 7.0 percent for type I and 14.0 percent for type II when tested as specified in 4.4.12.

3.4 Color.—The color of the talcum powder shall be white. When specified, the color shall match a sample provided by the procuring activity.

3.5 Odor.—The finished powder shall be mildly fragrant.

3.6 Fineness.—When tested as specified in 4.4.9, 98 percent of the finished talcum powder shall pass through a U. S. Standard Sieve No. 200.

3.7 Workmanship.—The finished product shall be a clean, uniform cosmetic powder having good covering power, slip and adher-

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ence to the skin. It shall be free from sharp particles and grit and shall conform to the level of quality established by this specification.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Inspection responsibility. — The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services or any other commercial laboratory acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction inspection.—When preproduction sample is specified, the sample shall be inspected and tested for conformance with the requirements of this specification.

4.3 Inspection for acceptance. — Inspection shall be in accordance with the provisions set forth in Military Standard MIL–STD–105 except where otherwise indicated hereinafter.

4.3.1 *Inspection of the end item.*

4.3.1.1 *Examination of the end item.*—The end item shall be examined in accordance with the classification of defects and acceptable quality levels (AQL's) set forth herein. The lot size shall be expressed in units of cans of talcum powder.

4.3.1.1.1 *Examination of preparation for delivery.*—An examination will be made to determine that packaging, packing and marking comply with the requirements of section 5 of this specification. Defects will be scored in accordance with the table below. The sample unit shall be one shipping container fully prepared for delivery. The lot size shall be the number of containers in the inspection lot.

Examine	Defect
Marking, exterior and interior	Incorrect; illegible; omitted; of improper size, location, sequence or method of application.
Materials	Any nonconforming component; component missing, damaged or otherwise defective affecting serviceability.
Workmanship	Inadequate application of components such as: Incomplete closure of case liners, container flaps, loose or inadequate strapping; Bulging or distortion of containers.
Contents	Number per container is more or less than required.

4.3.1.1.2 *Examination of the finished product.*—The sample unit for this examination shall be one filled container.

Examine	Defect
Finished product:	
Color	Not white. Does not match sample (when provided).
Odor	Not mildly fragrant. Objectionable.
Workmanship	Not clean. Not uniform. Does not exhibit characteristic slip. Does not adhere to skin in characteristic manner.

4.3.1.1.3 *Examination of the filled container for determining compliance with the*

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requirements for net contents.—The sample unit for this examination shall be one filled container.

Examine	Defect
Average net contents	Average net contents less than specified weight per container.

4.3.1.1.4 *Inspection levels and acceptable quality levels (AQL).*—Inspection levels and acceptable quality levels expressed in defects per hundred units, as applicable shall be as follows:

	Inspection level	*AQL*
For examination in 4.3.1.1.1	L–4	4.0
For examination in 4.3.1.1.2	L–1	2.5
For examination in 4.3.1.1.3	L–2	NA

4.3.2 *Testing of the end item.* — The end item shall be tested for the characteristics specified in table I. All requirements shall be applicable to the sample unit which shall be one 4-oz. can of talcum powder. One determination shall be made per sample unit. The lot size shall be expressed in units of cans of talcum powder. The inspection level shall be L–1 of Military Standard MIL–STD–105. The acceptable quality level shall be 4.0 defects per hundred units.

4.4 Tests:

4.4.1 *Starch.*—Warm a portion of the sample with distilled water. Cool and add a few drops of dilute (about 0.01*N*) iodine solution. A blue color indicates the presence of starch.

4.4.2 *Carbonates.*—Warm a portion of the sample with N/2 hydrochloric acid and observe solution for effervescence which indicates the presence of carbonates.

TABLE I.—*End item testing requirements*

Characteristic	Requirement reference	Test method reference	Results reported	
			Numerically to nearest	Pass or fail
Presence of starch	3.3	4.4.1	—	X
Presence of carbonate	3.3	4.4.2	—	X
Magnesium content	3.3.1	4.4.3	0.1 percent	—
Lead content	3.3.2	4.4.4	part/million	—
Arsenic content	3.3.2	4.4.5	part/million	—
Type II only Boric acid content	3.3.3	4.4.6	0.1 percent	—
Type II only Zinc stearate content	3.3.3	4.4.7	0.1 percent	—
pH	3.3.4	4.4.8	—	X
Fineness	3.6	4.4.9	1.0 percent	—
Presence of sharp particles	3.7	4.4.11	—	X
Presence of grit	3.7	4.4.10	—	X
Loss on ignition	3.3.5	4.4.12	0.17	

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4.4.3 *Magnesium oxide content.*—Proceed in accordance with United States Pharmacopoeia, test for identity. When completed, filter out precipitate, dry, weigh and calculate as magnesium oxide, M_gO.

4.4.4 *Lead content.*—Place 2 grams of sample in a platinum dish and incinerate for 1-2 hours at 550°-525°C. Cool and treat with 5 ml. of 60 percent $HClO_4$. Take down to fumes on a hot plate with 3 successive portions of HF. Cool and dilute with water.

(a) If the solution remains clear, transfer to a separatory funnel, dilute to 150-200 ml., add 10 grams of citric acid and proceed with the dithizone extraction of lead as given in the Methods of Analysis of the Association of Official Agricultural Chemists, finally estimating lead as directed.

(b) If the solution becomes turbid upon dilution, filter off as much as possible of the precipitate catching the filtrate in a 500-ml. glass-stoppered Erlenmeyer flask. Dilute to about 200 ml., add 10 grams of citric acid, and proceed with a sulfide separation of lead in accordance with the above publication using as little copper sulfide as required for the lead sulfide. Ignore colloidal material that may pass through the filter. Estimate as before with either the electrolytic procedure or the dithizone procedure.

4.4.5 *Arsenic content.*—Remove silica by a treatment similar to that used for lead but use a mixture of H_2SO_4 and HNO_3 (about 5 ml. of each) and take down to fumes before starting the HF volatilization. After cooling, dilute cautiously with water, make to volume, filter and use a suitable aliquot of the acid solution for determination of arsenic by the Gutzeit method. See methods of analysis of the Association of Official Agricultural Chemists.

4.4.6 *Boric acid content (type II only).*—Weigh 20,000 grams of sample into a platinum dish. Add sodium hydroxide solution until distinctly alkaline (to phenolphthalein indicator) then evaporate to dryness. Ignite until all organic matter is thoroughly charred, avoiding intense red heat; cool, digest with about 20 ml. of hot water and add dilute hydrochloric acid dropwise until distinctly acid. Filter into 100-ml. volumetric flask and wash with a little hot water. (Volume of filtrate should not exceed 50-60 ml.). Return filter containing any unoxidized carbon to platinum dish, make alkaline by wetting thoroughly with lime water, dry on steam bath, and ignite to a white ash. Dissolve ash in a few ml. of weak hydrochloric acid (equal volumes of dilute hydrochloric acid and water) and add to liquid in 100-ml. flask, rinsing dish with a few ml. of water. To combined solutions, add 0.5 gram of calcium chloride (anhydrous) and a few drops of phenolphthalein indicator, and 10-percent sodium hydroxide solution until a permanent light pink color is produced. Finally dilute to 100 ml. mark with lime water, mix, and filter through dry filter. To 50 ml. of filtrate add weak sulfuric acid (equal volumes of dilute sulfuric acid and water), until pink color disappears, then add methyl orange indicator and continue addition of the acid until yellow color is changed to pink. Boil about 1 minute to expel carbon dioxide. Cool, and carefully add 0.2*N* sodium hydroxide until liquid assumes yellow tinge, avoiding excess of alkali. (All of the boric acid is now in free state with no uncombined sulfuric acid present). Add a volume of glycerol (which is neutral to phenolphthalein) equal to volume of solution and a few drops of phenolphthalein indicator, note buret reading, and again titrate solution with the standard sodium hydroxide solution until pink color develops. Add a little more glycerol, and if pink color disappears continue addition of standard alkali until pink color reappears. Repeat alternate addition of glycerol and standard alkali until a permanent end point is reached. 1 ml. 0.2*N* sodium hydroxide equals 0.124 percent boric acid.

4.4.7 *Zinc stearate content (type II only).*

(1) *Total stearate.*—Weigh 20,000 grams of the powder into a 250-ml. beaker, and add 6 ml. of HCl, dilute. Leave on a steam bath for 10 minutes. Add 100 ml. of petroleum ether. Place the mixture under a stirring de-

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vice equipped with a glass stirrer, and stir rapidly for about 1 minute. Disconnect the stirrer from the machine, but leave it in the beaker. Decant the ether layer into a tared Pyrex dish, and place the dish on a steam bath. Repeat the extraction twice. (Evaporation of the first extract is so rapid that by the time the second extract is ready for decantation there will be room for it in the dish). Evaporate the combined extracts until bubbling has just ceased and a smooth layer of melted fatty acids remains in the dish. Remove the dish immediately, for fatty acids are volatile at the temperature of the steam bath. Cool, and weigh as total fatty acids.

$$\frac{100 \times \text{total fatty acids} \times 0.9683}{2}$$

= Percent of stearate as $(C_{17}H_{35}CO)_2O$

The addition of hydrochloric acid has a double purpose. It releases stearic acid for extraction and decomposes any starch present. Decomposition products of the starch pass into the filtrate obtained in 2, where they do not interfere.

The decantation involves no danger of mechanical loss, since the powder particles are tenaciously retained in the acid layer. The petroleum ether layer remains clear even during the stirring.

(2) *Total zinc oxide.*

To the acid layer remaining after decantation, add, in the order named, 6 ml. of NH_4OH, 50 ml. of Wulfing's precipitant for magnesium, and 50 ml. of 95 percent ethyl alcohol. Connect the glass stirrer to the stirring device, and stir for 5 minutes. Leave overnight, preferably in a refrigerator. (If magnesium is absent, an hour's standing will suffice). Filter through a fine 12½ cm. ashless paper, catching the filtrate in a 600-ml. beaker. (Filtration can be hastened by suction, but in this case the apex of the paper must be supported by a platinum cone). Wash the residue five times with a (1 + 1) mixture of Wulfing's precipitant and 95 percent ethyl alcohol. Neutralize the filtrate, and render it slightly acid with HCl. Transfer to a 500-ml. volumetric flask, make to the mark, and on a 200-ml. aliquot determine zinc as the pyrophosphate. (Scott: "Standard Methods of Chemical Analysis").

$$\frac{5 \times 100 \times 0.5340 \times \text{wt. zinc pyrophosphate}}{2 \times 2}$$

= Percent of zinc oxide

Wulfing's precipitant precipitates magnesium, calcium, bismuth, aluminum, and iron, but not zinc or boric acid. If boric acid is present it will contaminate the zinc phosphate precipitate. For removal of boron the filtrate is evaporated to dryness, methyl alcohol and sulfuric acid added, and the methyl borate boiled off.

4.4.8 *pH.*—Shake a portion of the sample thoroughly in distilled water until a suspension is formed. Test with both red and blue litmus papers.

4.4.9 *Fineness (coarse particles).*—Dry a No. 200 sieve in an oven at 103° to 110°C. Cool, and weigh. Place a 30 g. sample in a mortar, moisten with alcohol, and then add 100 ml. of water. Thoroughly mix by gentle pressure with the finger to break up all the lumps, and wash through the sieve with a gentle stream of water, stirring gently with a camel's hair brush until nothing more passes through the sieve. Dry the sieve at 105° to 110°C. for 1 hour; cool and weigh. From the increase in weight of the sieve and the weight of sample calculate the percentage of coarse particles. Save the residue for grit test paragraph below.

4.4.10 *Grit.*—Transfer the residue retained on the sieve (see 4.4.9) to a porcelain or glass plate. Examination with a steel spatula will indicate grit, if present.

4.4.11 *Sharp particles.*—Examine a portion of the sample under a microscope using a magnification of 200 diameters for the presence of sharp particles.

4.4.12 *Loss on ignition.*—Weigh accurately 1 gram of the test sample in a tared platinum crucible provided with cover. Gradually apply heat to crucible at first, then strongly ignite to constant weight.

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Percent loss on ignition =

$$\frac{\text{Weight loss}}{\text{Sample weight}} \times 100$$

5. PREPARATION FOR DELIVERY

(Federal Standard No. 102 shall be referred to for definitions and application of the various levels of packaging and packing for civilian procurements).

5.1 Packaging.—Packaging shall be level A or C as specified (see 6.2).

5.1.1 *Level A.* — One pound of talcum powder of one type only shall be packaged in a can conforming to type V, class 7 of Federal Specification PPP–C–96. Can closure shall be a revolving sifter type.

5.1.2 *Level C.*—Talcum powder shall be packaged in accordance with the industry's practice.

5.2 Packing.—Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 *Level A.*—Talcum powder of one type only, packaged as specified in 5.1, shall be packed as specified in the appendix of Federal Specification PPP–C–96 for overseas shipment.

5.2.2 *Level B.*—Talcum powder of one type only, packaged as specified in 5.1, shall be packed as specified in the appendix of Federal Specification PPP–C–96 for domestic shipment.

5.2.3 *Level C.*—Talcum powder, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Civil agencies.*—In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Federal Standard No. 123.

5.3.2 *Military requirements.*—In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Military Standard MIL–STD–129.

6. NOTES

6.1 Intended use.—Type I talcum powder is intended for general comfort use. Type II talcum powder is intended for use where a mildly acidic powder is required.

6.2 Ordering data. — Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number and date of this specification.
(*b*) Type of talcum powder required (see 1.2).
(*c*) Whether a pre-production sample is required (see 3.1).
(*d*) Whether a color sample will be furnished (see 3.4).
(*e*) Inspection method and tests (see 4.1).
(*f*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded, by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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